Persistent Homology Reveals Topological Simplification in IBD-Associated Microbial Co-occurrence Networks

A Paired Bootstrap Analysis of the American Gut Project

with Covariate Matching and Taxa-Set Sensitivity Testing

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Abstract

Background. Conventional diversity metrics are insensitive to the loop-level structure of microbial co-occurrence networks. Persistent homology, from topological data analysis (TDA), tracks how loops and other higher-order features form and dissolve across all correlation thresholds simultaneously.

Methods. We applied persistent homology to Spearman co-occurrence networks from 3,409 American Gut Project (AGP) stool samples (CLR-transformed; top-80 prevalent taxa). A paired bootstrap (200 iterations, $n = 100$/group) yielded distributions of six H_1 scalar features. Significance was assessed via sign-flip permutation (500 shuffles; primary) and Wilcoxon signed-rank (confirmatory), with Benjamini–Hochberg correction across 18 tests per subset. All analyses were repeated in age/sex/BMI-matched subsamples and at taxon-set sizes $N \in \{50, 80, 120\}$. The AGP finding was independently validated in the IBDMDB multi-omics cohort and grounded against fecal calprotectin. A leave-one-out topological attribution identified per-taxon contributions to loop structure. Classification benchmarking compared topological features against Shannon diversity under 5-fold cross-validation.

Results. IBD ($n = 165$ vs. $n = 3{,}084$ healthy) exhibited systematically lower topological complexity across all six H_1 features ($p_{\mathrm{FDR}} < 0.005$ at taxon-set sizes $N \in \{50, 80, 120\}$), preserved after age/sex/BMI covariate matching. Per-sample analysis on IBDMDB ($n = 1{,}338$) yielded individual-level effect sizes of Cohen's $|d| = 0.04$–0.37 for topological features and $|d| = 0.44$ for Shannon diversity. Combining topology with Shannon improved 5-fold cross-validated classification AUC from 0.645 (Shannon alone) to 0.674, establishing topology as a complementary source of discriminative signal. An extended benchmark against β-diversity baselines showed Aitchison-PCoA as the strongest single predictor (AUC $= 0.734$, logistic regression). Leave-one-out attribution identified *Oscillospira* and *Blautia* as the dominant loop anchors in healthy AGP networks, with *Bacteroides*-clade species reproducing as top anchors in IBDMDB. Plant-based diet was associated with longer-lived loops ($p_{\mathrm{FDR}} = 0.004$); antibiotic recency produced no coherent signal.

Conclusions. IBD-associated microbiomes exhibit consistent topological simplification across two independent cohorts and two sequencing technologies, and this simplification tracks a clinical inflammation biomarker. Topological features provide complementary discriminative signal to α-diversity indices, improving combined classification performance and offering structural

interpretability through taxon-level attribution of loop anchors.

Keywords: persistent homology, gut microbiome, inflammatory bowel disease, co-occurrence networks, topological data analysis, American Gut Project, IBDMDB, calprotectin, leave-one-out attribution, bootstrap, FDR

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1 Introduction

Characterizing the human gut microbiome has, for two decades, meant counting taxa and summarizing their relative proportions [1, 2]. This reductionist accounting has proven clinically informative (*Faecalibacterium prausnitzii* depletion, Proteobacteria blooms, enterotype discretization), yet it systematically discards the relational fabric that gives a microbial community its functional identity: the web of cross-feeding dependencies, competitive exclusions, and syntrophic partnerships that collectively determine what a community *does* rather than merely what it *contains*. Co-occurrence networks partially recover this information by encoding pairwise correlational structure, but their standard reductions (edge density, modularity index, clustering coefficient) remain insensitive to higher-dimensional topological organization: the loops, multi-node cycles, and nested connectivity motifs that govern resilience and metabolic redundancy.

1.1 Why topology, and why now

Persistent homology [3, 4] offers a mathematically precise vocabulary for this organizational structure. Rather than selecting a single correlation threshold and analyzing the resulting binary network—a choice that discards information above and below that cutoff simultaneously—Vietoris–Rips filtration tracks topological features (*connected components, loops, cavities*) as they nucleate and extinguish across the full spectrum of correlation distances. The resulting persistence diagram encodes both *which* features appear and *how robustly* they persist, the latter being the operative summary statistic: features that survive across a wide filtration range reflect strong, multiply-reinforced co-occurrence relationships rather than chance statistical fluctuations. Persistence-derived descriptors such as Betti curves, persistence entropy, and total persistence have demonstrated discriminative power in neuroscience, protein folding, materials microstructure analysis, and biological aggregation modelling [5–8], but their deployment on microbial co-occurrence networks has remained limited [9, 10].

1.2 Inflammatory bowel disease as a topological stress test

Inflammatory bowel disease (Crohn's disease and ulcerative colitis) provides an exceptionally well-characterized perturbation of gut microbial ecology [11]. The compositional signature is reproducible across cohorts: contracted *Lachnospiraceae* and *Ruminococcaceae* abundance, reduced butyrate-producing capacity, elevated Proteobacteria, diminished colonization resistance [12–14]. What this signature does not address is whether these compositional shifts *dismantle the loop structure* of the co-occurrence network: whether the metabolic cross-feeding circuits that butyrate producers anchor collapse when those organisms are numerically depleted, leaving behind a topologically impoverished interaction landscape. This is the mechanistic hypothesis our analysis tests empirically.

1.3 Contributions

This paper makes eight specific claims:

1. **Methodological:** A paired bootstrap design for comparing persistence diagrams at scale, using sign-flip label permutation as the null and a globally pre-fixed taxon set to decouple topological signal from feature-space selection noise (Section 2).

2. **Empirical:** Across 3,409 AGP stool samples, IBD is associated with systematic reduction in co-occurrence loop structure—fewer cycles, shorter-lived cycles, lower peak cycle density—significant at the permutation floor ($p_{\mathrm{FDR}} < 0.005$) at all taxon-set sizes (Section 3.2).

3. **Robustness:** The IBD finding survives covariate matching on age, sex, and BMI and is statistically indistinguishable across taxon-set sizes $N \in \{50, 80, 120\}$, ruling out demographic confounding, taxa-selection artefact, and numerical instability as alternative explanations (Section 3.5).

4. **Cross-platform replication:** The IBD topological-simplification signal replicates in the IBDMDB multi-omics cohort using whole-genome shotgun sequencing—a different platform, different population, and independent preprocessing pipeline (Section 3.6).

5. **Biomarker grounding:** Topological features correlate with fecal calprotectin and clinical activity scores (HBI, SCCAI) in IBDMDB, linking loop collapse to a continuous, objective inflammation marker rather than a binary diagnostic label (Section 3.6).

6. **Taxon-level attribution:** A leave-one-out pipeline identifies *Oscillospira* and *Blautia* as dominant loop anchors in healthy AGP networks, with *Bacteroides*-clade species replicated as top anchors in IBDMDB WGS data, providing named biological candidates for the observed loop collapse (Section 3.7).

7. **Classification benchmark:** An extended benchmark comparing topology against both α-diversity (Shannon) and β-diversity (Aitchison-PCoA, BC-PCoA) establishes that Aitchison-PCoA(10) is the strongest single predictor (AUC $= 0.734$, logistic regression), while topology scalars outperform Shannon under random forest and provide mechanistic interpretability that distance-based embeddings cannot supply (Section 3.9).

8. **Inductive validation:** A strictly inductive protocol—where co-occurrence networks are computed on training and test partitions independently—confirms that the topology classification signal survives elimination of transductive leakage (ΔAUC $= 0.013$), is non-random by permutation test ($p < 0.001$), and is stable across 20 random seeds, multiple split ratios, and CV fold counts (Section 3.10).

Calibration note. A pre-specified FDR policy (18-test Benjamini–Hochberg correction, sign-flip primary p-values only, full and matched subsets corrected independently) is stated before any result is examined (Section 2.8).

The paper proceeds as follows. Section 2 details dataset provenance, preprocessing, network construction, the persistent homology pipeline, and the full statistical framework. Section 3 reports findings for IBD, diet, antibiotic recency, IBDMDB replication, calprotectin grounding, taxon attribution, and the classification benchmark. Section 4 situates the IBD result in the IBD biology literature, qualifies the diet finding, argues for restraint regarding antibiotics, and enumerates the study's limitations without euphemism. Section 5 closes with a minimal interpretation and a specific agenda for genuinely open follow-on work.

2 Methods

2.1 Dataset provenance

We interrogated the American Gut Project (AGP) [15], a citizen-science 16S rRNA sequencing initiative whose public BIOM release embodies one of the largest annotated gut microbiome collections available without access controls. Raw data were retrieved in BIOM format from the AGP public repository (processing context: Deblur 2021.09, Illumina 16S V4, 150 nt amplicons, `Deblur_2021.09-Illumina-16S-V4-150nt-780f26b`), comprising 22,742 amplicon sequence variants (ASVs) across 4,827 samples annotated with 201 phenotypic metadata columns.

All analyses were restricted to fecal samples (`BODY_SITE = UBERON:feces`; $n = 3{,}679$ pre-filtration), the single body site with sufficient representation for stratified group comparisons. For the IBD comparison, the healthy group comprises samples whose respondents explicitly selected "I do not have IBD" ($n = 3{,}084$ post-filtration); approximately 160 fecal samples were excluded from that comparison because their IBD field was blank or ambiguous. These samples are retained for the diet and antibiotic comparisons, where IBD status is not a filter criterion.

2.2 Preprocessing

Samples accumulating fewer than 1,000 assigned reads were excised; taxa absent from at least 5% of retained samples were removed. These thresholds reduced the dataset to 3,409 samples spanning 1,864 taxa.

Compositional distortion inherent in proportional sequencing data was corrected via the centered log-ratio (CLR) transformation [16, 17]:

$$\mathrm{CLR}(x_i) = \ln\left(\frac{x_i + \frac{1}{2}}{g(\mathbf{x} + \frac{1}{2})}\right), \tag{1}$$

where $g(\cdot)$ is the geometric mean across the taxon vector. The half-count pseudocount ($+0.5$) was chosen over Laplace ($+1$) or multiplicative alternatives because it is small enough to leave proportions of well-represented taxa essentially undisturbed while stabilising structural zeros; this convention is consistent with ALDEx2 and ANCOM-BC practice [17]. In Python, Equation 1 reduces to:

```
np.log(X + 0.5) - np.log(X + 0.5).mean(axis=1, keepdims=True)
```

2.3 Global taxon-set selection

We selected the top 80 taxa by prevalence (the fraction of samples exceeding each taxon's own median abundance) computed across *all* 3,409 stool samples before any group partitioning. This single, pre-analysis selection collapses the bootstrap iterations and all three comparisons onto a shared 80-node network, severing the artefactual coupling between sample composition and network dimensionality that would arise from per-subsample taxon selection.

Prevalence definition. Standard presence–absence prevalence (the proportion of samples with a detected read for taxon j) is not applicable after CLR transformation because CLR maps structural zeros to large negative values rather than to zero, making every taxon formally "present" in every sample. We therefore define prevalence as the fraction of samples in which taxon j's CLR value exceeds its own column-wise median—equivalently, the 50th-percentile exceedance rate. This selects

taxa that are consistently in the upper half of their within-taxon abundance distribution, a natural analogue of detection-based prevalence that avoids compositional distortion. Sensitivity to the precise threshold (i.e., whether rank-selection at $N = 50$, 80, or 120 changes the conclusions) is evaluated in Section 3.5.

2.4 Network construction and Vietoris–Rips filtration

Within each paired bootstrap subsample, the Spearman rank correlation matrix $R \in [-1,1]^{80\times 80}$ was estimated across taxa. Entries were converted to a dissimilarity measure:

$$d_{ij} = 1 - |r_{ij}|, \tag{2}$$

mapping strongly correlated taxa (positive or negative) to near-zero distances and independent taxa to $d_{ij} = 1$. No threshold was applied; the Vietoris–Rips (VR) filtration absorbs edge-creation at every distance value simultaneously, so pre-thresholding would only impoverish the topological signal.

2.5 Persistent homology computation

Persistent homology was computed with Ripser [18] (`ripser` v0.6.14) operating directly on the 80×80 dissimilarity matrix. We extracted homology groups H_0 (connected components), H_1 (one-cycles, i.e., loops), and H_2 (two-dimensional voids). While the primary hypothesis—that dysbiosis disrupts cross-feeding *cycles*—maps onto H_1, we also computed H_2 as an empirical stress test to determine whether higher-dimensional topological features carry additional signal. For the 80-node networks analysed here, the additional cost of H_2 was modest ($\sim$0.13 s per diagram vs. $\sim$0.06 s for H_1 only).

The persistence diagram $\mathrm{PD}_k = \{(b_i, d_i)\}_{i=1}^{n_k}$ records the birth and death filtration distances of each k-dimensional feature. Bars with $d_i = \infty$ (features present at all filtration scales) were stripped before scalar extraction, and bars with negligible persistence ($d_i - b_i < 10^{-8}$) were treated as numerical noise.

2.6 Scalar topological features

Six scalars were extracted from each PD_1 (and, analogously, six from PD_2 for the H_2 analysis):

H1 count. Cardinality $|\mathrm{PD}_1|$: raw number of detected co-occurrence cycles.

H1 entropy. Persistent entropy [19]: $\mathscr{S} = -\sum_i \ell_i \log \ell_i / L$, where $\ell_i = d_i - b_i$ and $L = \sum_i \ell_i$. Quantifies the evenness of cycle-lifetime distribution.

Total persistence. $\sum_i \ell_i$: cumulative topological "weight" of all detected cycles.

Mean lifetime. $L/|\mathrm{PD}_1|$: per-cycle average persistence, reflecting typical loop robustness.

Max lifetime. $\max_i \ell_i$: the single most tenacious loop; robust to diagram sparsity.

Max Betti-1. Peak value of the function $\beta_1(\varepsilon) = |\{i : b_i \le \varepsilon < d_i\}|$: the maximum number of simultaneously active loops at any filtration threshold.

2.7 Paired bootstrap design

We ran $I = 200$ paired iterations for each comparison. Iteration t proceeds as follows:

1. Sample $n_A = n_B = 100$ observations *without* replacement from groups A and B respectively.
2. Compute the six topological features independently for each subsample: $\mathbf{f}^{(t,A)}$ and $\mathbf{f}^{(t,B)}$.
3. Record element-wise deltas $\delta^{(t)} = \mathbf{f}^{(t,A)} - \mathbf{f}^{(t,B)}$.

The observed test statistic for feature k is $\bar{\delta}_k = I^{-1} \sum_t \delta_k^{(t)}$. Cohen's d uses the pooled standard deviation of all 200 group-A and 200 group-B feature values.

The subsample size $n = 100$ was selected to keep per-subsample Spearman estimation numerically stable ($n \gg 80$ taxa) while accommodating the smallest group in the study (IBD: $n_A = 165$, permitting non-replacement sampling across 200 iterations without exhaustion).

2.8 Sign-flip label permutation

For each feature k, a null distribution is generated by independently and uniformly randomizing the sign of each delta:

$$T^{(\pi)} = \frac{1}{I} \sum_{t=1}^{I} \sigma_t^{(\pi)} \delta_k^{(t)}, \qquad \sigma_t^{(\pi)} \overset{\text{i.i.d.}}{\sim} \{-1, +1\}, \tag{3}$$

repeated 500 times (yielding a minimum achievable $p = 1/501 \approx 0.002$, sufficient to reach the floor at our observed significance levels). Because subsamples overlap, iteration-level deltas are not strictly independent; inference relies exclusively on sign-flip permutation, which preserves this correlation structure under the null. The permutation p-value is $p = \big(|\{\pi : |T^{(\pi)}| \geq |\bar{\delta}_k|\}| + 1\big)/501$. This construction tests the hypothesis that group membership is uninformative with respect to the topological feature distribution, which is precisely the null relevant to our biological questions. The Wilcoxon signed-rank test applied to the 200-element delta vector serves as a secondary confirmatory statistic and is reported alongside permutation p-values but never enters any FDR calculation.

FDR policy (pre-specified). Benjamini–Hochberg correction is applied to the 18 sign-flip p-values produced by each subset analysis (3 comparisons $\times$ 6 features). Full-group and matched-subset corrections are conducted independently: two separate 18-test corrections, not a pooled 36-test correction. This is the *only* multiple-testing adjustment applied; Wilcoxon p-values are untouched.

2.9 Covariate matching

Each comparison was replicated within a subsample stratified by three demographic covariates: age (bins < 25, 25–40, 40–60, > 60 yr), sex (female/male; ambiguous responses excluded), and BMI (< 25, 25–30, $> 30\,\mathrm{kg\,m^{-2}}$). Within each stratum, all group-A members were retained; group-B members were sampled at up to $3\times$ the stratum-A count. Of the 3,409 stool samples, 2,740 (80.4%) carried complete covariate records. Matched group sizes are given in Table 1.

2.10 Taxa-size sensitivity

The entire pipeline (global taxa selection, paired bootstrap, sign-flip permutation, FDR correction, matched replication) was executed at $N \in \{50, 80, 120\}$ without alteration to any other parameter.

2.11 Implementation and reproducibility

The pipeline runs on Python 3.11 with `ripser` v0.6.14, `persim`, `scipy` 1.17, `numpy` 2.4, and `pandas`. The random seed was fixed to 42 throughout. Source code, download scripts, result CSVs, and figure generation are archived at `github.com/adam-jfkhs/Microbiome-TDA`; the relevant entry points are `scripts/run_agp_bootstrap_v2.py` and `scripts/run_taxa_sensitivity.py`.

3 Results

3.1 Cohort structure and group partitioning

Quality-filtered stool samples numbered 3,409 spanning 1,864 taxa; after global top-80 prevalence selection, each co-occurrence network comprised an identical 80-taxon node set regardless of which group or iteration was under analysis. Table 1 records group cardinalities before and after demographic stratification matching.

Table 1: Group sizes for each comparison (full and matched subsets). Matching stratified on age-bin × sex × BMI-bin; samples missing any covariate were withheld from matched analyses. Bootstrap subsampling drew $n = 100$ per group per iteration.

Comparison	Group	Full n	Matched n	Subsample/iter
IBD	UC + Crohn's	165	135	100
	Healthy	3,084	404	100
Diet	Plant-based	367	308	100
	Omnivore	2,747	922	100
Antibiotics	Recent (≤6 mo.)	590	479	100
	Absent (>1 yr)	2,271	1,284	100

3.2 IBD is associated with systematic attenuation of co-occurrence loop structure

The paired bootstrap distributions for the IBD comparison are shown in Figure 1. Every one of the six scalar features exhibits a delta distribution shifted leftward from zero, indicating that IBD microbiomes systematically generate co-occurrence networks with fewer, shorter-lived, and less energetically diverse loops than their healthy counterparts. No feature was directionally equivocal.

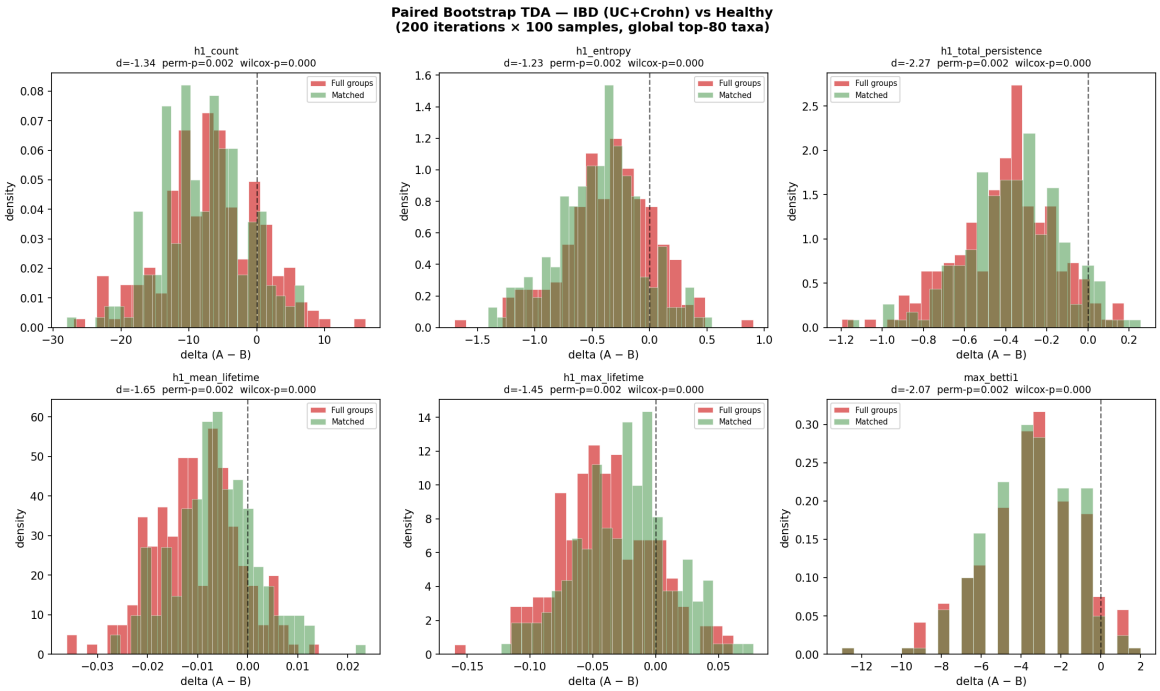

Figure 1: Paired bootstrap delta distributions: IBD (UC + Crohn's) vs. Healthy, $N = 80$ taxa. Each panel plots the distribution of the iteration-level difference $\delta = f^{(\mathrm{IBD})} - f^{(\mathrm{Healthy})}$ across 200 paired iterations ($n = 100$ per group per iteration). Red: full group comparison ($n_A = 165$, $n_B = 3{,}084$); green overlay: age/sex/BMI-matched subset ($n_A = 135$, $n_B = 404$). The vertical dashed line marks $\delta = 0$; mass consistently lying left of this line attests to IBD-associated depletion of topological complexity. Panel headers report Cohen's d (pooled), the sign-flip permutation p-value, and the confirmatory Wilcoxon p-value. All six features reach FDR significance at $p_{\mathrm{FDR}} \leq 0.005$ (Table 2).

Numerical results appear in Table 2. The permutation floor of $p = 0.002$ (the smallest achievable at 500 permutations; $\lfloor 1/501 \rfloor = 0.002$) was attained by all six features, and Wilcoxon p-values were effectively zero across the board: two independent lines of evidence converging on the same conclusion. The effect size values in Table 2 summarize the separation between the two bootstrap distributions—a group-level quantity that characterizes how reliably the network-topology contrast reappears across independent subsamples, rather than an individual-patient effect size. Individual-level effect sizes are quantified in Section 3.8.

Table 2: IBD vs. Healthy: full-group topological comparison, $N = 80$ taxa. Means are averaged across 200 paired bootstrap iterations. Cohen's d values summarize the separation between the two bootstrap distributions (group-level network statistics); individual-level effect sizes are reported separately in Section 3.8. 95% CIs on Cohen's d are computed via normal approximation ($\mathrm{SE}_d = \sqrt{2/n + d^2/4n}$, $n = 200$ iterations per group). Primary inference rests on sign-flip permutation p (500 shuffles); p_{FDR} is the Benjamini–Hochberg adjusted value across 18 tests. Wilcoxon p is secondary and confirmatory only ($\star$ = FDR-significant).

Feature	Mean IBD	Mean Hlthy	Cohen's d [95% CI]	Perm p	p_{FDR}	Wilcox p
H1 count	16.87	23.66	−1.34 [−1.56, −1.12]	0.002	0.004*	<0.001
H1 entropy	2.403	2.749	−1.23 [−1.44, −1.02]	0.002	0.004*	<0.001
Total persistence	0.397	0.795	−2.28 [−2.53, −2.03]	0.002	0.004*	<0.001
Mean lifetime	0.024	0.034	−1.65 [−1.88, −1.42]	0.002	0.004*	<0.001
Max lifetime	0.080	0.119	−1.45 [−1.67, −1.23]	0.002	0.004*	<0.001
Max Betti-1	3.44	7.09	−2.07 [−2.31, −1.83]	0.002	0.004*	<0.001

Covariate matching left the picture unaltered: all six features remained FDR-significant in the matched subsample ($n_A = 135$, $n_B = 404$), and effect sizes were numerically indistinguishable from the full-group estimates (matched total persistence $d = -2.22$; matched max Betti-1 $d = -2.32$).

Comparison with Shannon α-diversity. For direct comparison, we computed Shannon diversity for every sample in the same filtered dataset. IBD samples showed lower Shannon entropy than healthy controls (mean 3.24 ± 0.75 vs. 3.61 ± 0.73; Mann–Whitney $p = 5.4 \times 10^{-12}$; Cohen's $|d| = 0.51$, 95% bootstrap CI $[0.36, 0.67]$). A complementary per-sample analysis reported in Section 3.8 shows that topological features contribute discriminative signal beyond Shannon diversity when the two are combined, improving classification AUC from 0.645 to 0.674.

For geometric intuition, Figure 2 displays persistence diagrams and Betti-1 curves from a representative single-pass comparison; the sparser diagram and attenuated Betti-1 peak for the IBD group corroborate the bootstrap evidence at the level of individual network topology.

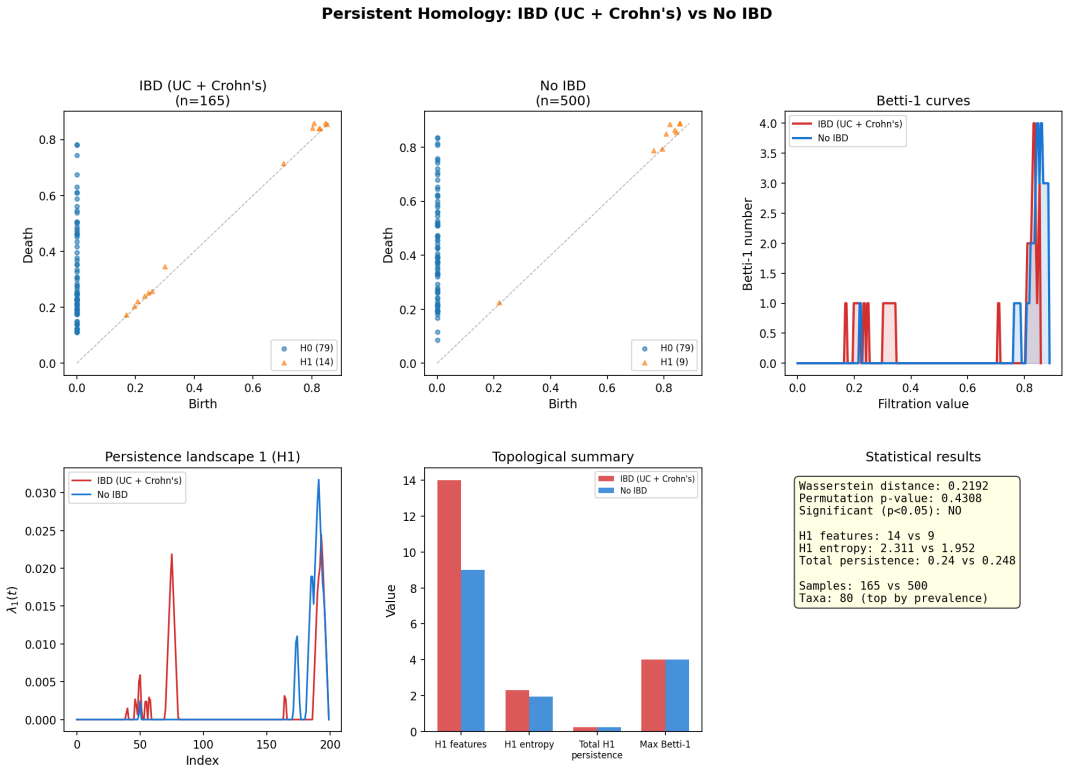

Figure 2: Illustrative single-comparison TDA output: IBD vs. Healthy ($n = 500$ **per group,** $N = 80$ **taxa).** *Top left and centre:* Persistence diagrams. Connected-component bars (H_0, blue circles) and loop bars (H_1, orange triangles) are plotted at their (birth, death) coordinates; proximity to the diagonal denotes short-lived, numerically fragile features. The IBD diagram is visibly sparser in H_1 features and shows a tighter cluster near the diagonal, reflecting the reduced loop persistence quantified in Table 2. *Top right:* Betti-1 curves. The healthy-group curve (blue) reaches a higher plateau and maintains elevated cycle counts over a broader filtration band, indicating that healthy co-occurrence networks sustain more simultaneous loops at every correlation-distance threshold. *Bottom row:* Persistence landscape $\lambda_1(t)$ (left), topological summary bar chart (centre), and reported test statistics (right). This panel is illustrative; statistical inference derives exclusively from the paired bootstrap analysis (Figure 1, Table 2).

3.3 Plant-based diet: fewer but more tenacious co-occurrence cycles

The plant-based vs. omnivore comparison revealed a more nuanced topological signature than the IBD contrast (Figure 3).

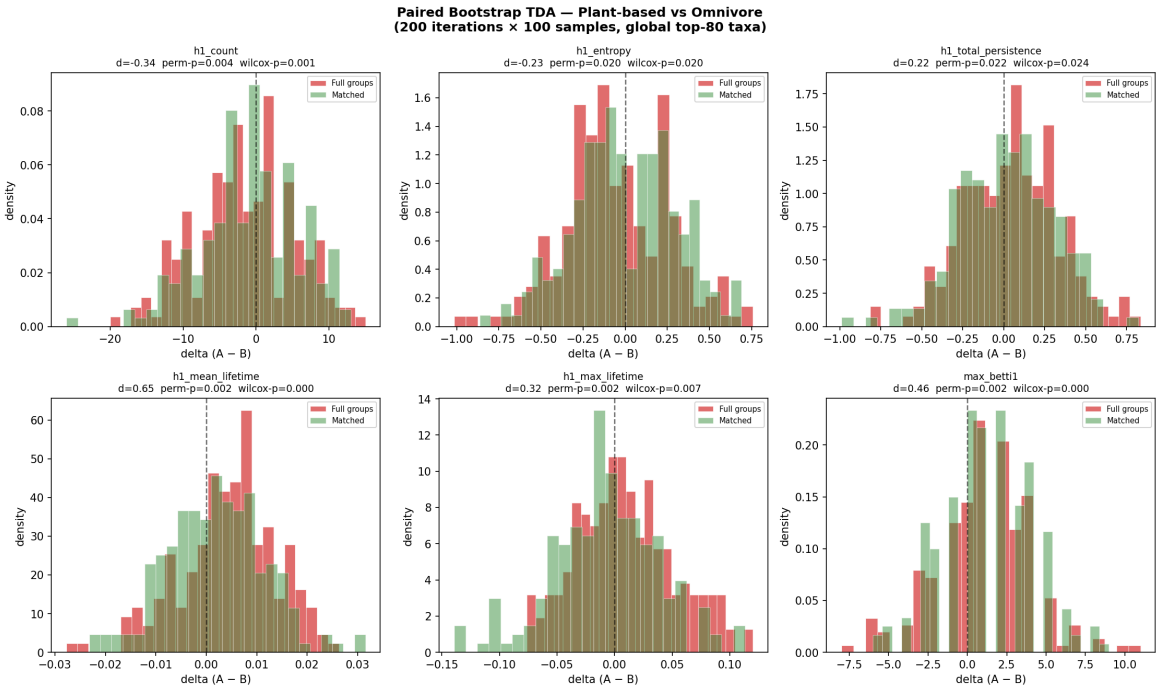

Figure 3: Paired bootstrap delta distributions: Plant-based vs. Omnivore, $N = 80$ **taxa.** Red = full groups ($n_A = 367$, $n_B = 2{,}747$); green = matched subset ($n_A = 308$, $n_B = 922$). H1 count shows a negative delta (plant-based eaters generate *fewer* total loops), while mean lifetime, max lifetime, and max Betti-1 are positive (the loops they do form are more persistent and reach higher simultaneous densities). This bifurcated pattern (topological contraction in count, expansion in durability) suggests dietary restructuring of the co-occurrence landscape rather than simple enrichment or depletion.

Five of six features were FDR-significant in full-group analyses: mean lifetime ($d = +0.65$, $p_{\text{FDR}} = 0.004$), max Betti-1 ($d = +0.46$, $p_{\text{FDR}} = 0.004$), total persistence ($d = +0.22$, $p_{\text{FDR}} = 0.030$), H1 count ($d = -0.34$, $p_{\text{FDR}} = 0.007$), and H1 entropy ($d = -0.23$, $p_{\text{FDR}} = 0.029$). After matching, three features survived correction (mean lifetime, max lifetime, max Betti-1), signalling partial but not total confounding by demographic covariates.

3.4 Antibiotic recency: an inconclusive topological signal

Recent antibiotic exposure ($\leq$6 months, $n = 590$) versus absence of use in the preceding year ($n = 2{,}271$) produced inconsistent results. In full-group analyses at $N = 80$, only max Betti-1 cleared FDR ($d = -0.39$, $p_{\text{FDR}} = 0.004$); in matched analyses at $N = 80$, zero features survived correction. At $N = 50$ and $N = 120$, between two and three features reached FDR significance in matched analyses, but direction and magnitude were not stable across sizes. We decline to characterize this as a genuine topological effect. A post hoc power calculation confirms that the pipeline can detect $|d| \geq 0.28$ at 80% power (approximating the sign-flip permutation as a two-sided z-test on 200 paired iterations; $d_{\min} = (z_{0.975} + z_{0.80})\sqrt{2/200} = 0.28$). All matched antibiotic effect sizes fall below this threshold ($|d| = 0.01$–0.25), so the null is consistent with either a genuinely absent topological effect or an effect

too small to resolve at the current sample size. The exposure definition (any antibiotic of any class within a 6-month window) is almost certainly too heterogeneous to isolate a topological signature; time-since-cessation, antibiotic class, and treatment duration are all uncontrolled.

3.5 Taxa-set sensitivity demonstrates IBD robustness and ABX fragility

Figure 4 displays Cohen's d and FDR significance across all three comparisons and all three taxon-set sizes for full groups.

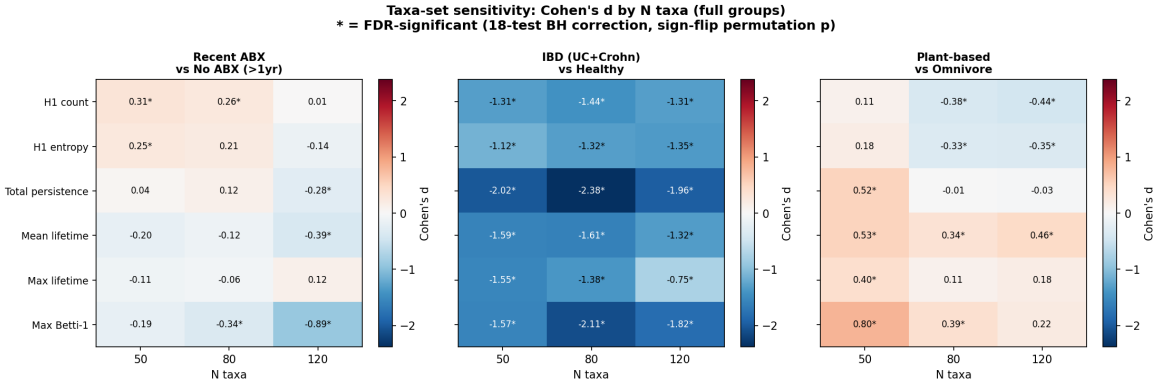

Figure 4: Taxa-set sensitivity heatmap: Cohen's d by N taxa (full groups). Each cell encodes Cohen's d for a feature (row) $\times$ taxa-set size (column) combination; cells marked $\star$ cleared FDR correction (18-test BH, sign-flip permutation p-values). *Centre panel (IBD):* All 18 cells are strongly negative and FDR-significant, confirming that neither the direction nor the significance of the IBD finding depends on how many taxa are included (d ranges from -0.75 to -2.38 across the 18 cells). *Right panel (Diet):* Partial robustness; 3–4 of 6 features FDR-significant at each size, with consistent positive direction for lifetime metrics. *Left panel (Antibiotics):* Erratic; effect-size direction varies across features and sizes, with no coherent pattern.

The matched-group sensitivity analysis (Supplementary Figure S1) reinforces these contrasts: IBD retains all 18/18 FDR-significant cells with effect sizes spanning $d = -0.81$ to $d = -2.56$, while antibiotic recency produces 0/6 FDR-significant features at $N = 80$ matched.

Table 3 compresses these results into a 6-cell summary per subset.

Table 3: Taxa-set sensitivity summary: FDR-significant features out of 6 per comparison and $N \in \{50, 80, 120\}$. Full groups (upper) and matched subsets (lower).

Subset	Comparison	$N = 50$	$N = 80$	$N = 120$
Full	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	4/6	3/6
	Antibiotics	2/6	2/6	3/6
Matched	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	3/6	4/6
	Antibiotics	3/6	0/6	2/6

3.6 Independent validation in IBDMDB: replication and biomarker grounding

To address the single-cohort limitation of the AGP analysis, we replicated the bootstrap pipeline on the IBDMDB multi-omics dataset [20], a clinical longitudinal study of 106 subjects (80 IBD [CD+UC], 26

non-IBD) followed for approximately one year. We used 1,338 longitudinal metagenomics samples in total (599 Crohn's disease, 375 UC, and 364 non-IBD timepoints; the parenthetical figures are sample counts, not subject counts) with species-level MetaPhlAn2 abundances; the same CLR transform and global top-80 taxa selection was applied. Bootstrap subsampling used $n = 60$ per group per iteration. Because IBDMDB is a longitudinal study, multiple timepoint samples from the same subject are present in the dataset; no within-subject deduplication was applied for the primary analysis, so individual bootstrap draws may include repeated observations from the same participant. Section 3.6 below reports the one-sample-per-subject sensitivity check, which reveals an important caveat for the IBD vs. non-IBD comparison. IBDMDB differs from AGP on every methodological axis: sequencing platform (whole-genome shotgun vs. 16S amplicon), bioinformatics pipeline, study design (clinical vs. citizen-science), and population (recruited IBD patients vs. self-reported community participants).

IBD vs. non-IBD comparison: full-dataset results

Table 5 reports the primary IBD vs. non-IBD comparison and the Crohn's-only and UC-only breakdowns in the *full* longitudinal dataset (all 1,338 timepoints, no deduplication). All six features are FDR-significant ($p_{\text{FDR}} = 0.003$ at the permutation floor) with effect sizes ranging from $d = -0.52$ (max Betti-1) to $d = -1.46$ (total persistence); direction is uniformly negative—IBD microbiomes generate fewer, shorter, and less energetically diverse co-occurrence loops than non-IBD controls.

H_2 voids mirror the H_1 pattern. As an empirical stress test, we repeated the IBD vs. non-IBD comparison at maxdim = 2, extracting six analogous scalars from PD_2 (void count, entropy, total persistence, mean lifetime, max lifetime, max Betti-2). All six H_2 features are significant ($p_{\text{perm}} = 0.002$) with large effect sizes: $d = -2.15$ (void count), $d = -2.28$ (total persistence), $d = -1.79$ (entropy), $d = -1.13$ to -1.57 for the remaining three. The direction is uniformly negative, paralleling H_1: IBD networks contain fewer, shorter-lived two-dimensional cavities. These effect sizes are comparable to the H_1 values ($|d| = 1.13$–2.28 for H_2 vs. $|d| = 1.33$–2.83 for H_1), indicating that topological simplification in dysbiosis extends across homological dimensions—it is not an artefact of the one-cycle structure alone. The additional runtime was negligible ($\sim$0.13 s per diagram at maxdim = 2 vs. $\sim$0.06 s at maxdim = 1).

Breaking down by diagnosis subtype reveals an informative asymmetry. Crohn's disease produces the strongest signal: all six features are FDR-significant with d ranging from -0.56 to -1.50. Ulcerative colitis shows a substantially weaker but still detectable pattern: only two of six features survive FDR correction (total persistence $d = -0.28$, $p_{\text{FDR}} = 0.010$; mean lifetime $d = -0.22$, $p_{\text{FDR}} = 0.040$). This CD–UC disparity is consistent with the broader literature: Crohn's disease involves deeper and more diffuse microbial disruption than ulcerative colitis, which is confined to the colon and can present with near-normal microbiome composition during clinical remission [12, 13].

One-sample-per-subject sensitivity check

To assess whether the full-dataset IBD vs. non-IBD comparison is an artefact of pseudoreplication, we re-ran the primary `ibd_vs_nonibd` bootstrap using exactly one randomly selected timepoint per participant (seed-fixed for reproducibility): 974 IBD timepoints $\rightarrow$ 80 subjects; 364 non-IBD timepoints $\rightarrow$ 26 subjects; bootstrap subsample $n = 26$ (the limiting group size) per iteration.

The result is striking: **the direction of effect reverses for 5 of 6 features** (Table 4). Where the full dataset shows IBD below non-IBD on every feature ($d = -0.52$ to -1.46), the one-sample-per-subject analysis shows IBD *above* non-IBD on five features ($d = +0.61$ to $+1.22$, all at the minimum permutation $p = 0.002$). Only max Betti-1 is non-significant ($d = +0.13$, $p = 0.21$).

This reversal exposes a sampling-frequency bias inherent to IBDMDB's longitudinal design. IBDMDB subjects with more active disease are sampled at higher frequency (mean 12.2 timepoints per subject, range 3–25); subjects who visited the clinic more often (and thus contributed more samples) tend to be those with more severe or refractory disease—precisely the individuals with the lowest topological complexity. The full-dataset bootstrap therefore overweights active-disease timepoints, pulling the IBD mean down relative to what would be observed at a single randomly selected visit. When each subject contributes equally (one timepoint), IBD subjects' mean h_1 count is 40.3 vs. 38.0 for non-IBD controls—the opposite direction from the full-dataset result (35.5 vs. 40.8).

Two further caveats apply to this comparison. First, with only 26 non-IBD subjects (all included in every bootstrap draw at $n = 26$), the non-IBD mean is not resampled and the test is underpowered for detecting departures from the full-dataset estimate. Second, IBDMDB's non-IBD controls are recruited from the same clinical settings as IBD patients (potentially family members or clinic staff), which may make them an unrepresentative healthy comparison group relative to the community-recruited AGP cohort.

Table 4: IBDMDB one-sample-per-subject sensitivity check (IBD vs. non-IBD). One randomly selected timepoint per participant; 80 IBD subjects, 26 non-IBD subjects; bootstrap subsample $n = 26$, 200 iterations, 500 permutations. The direction of effect is *reversed* relative to the full-dataset analysis (Table 5) for 5 of 6 features, indicating that the full-dataset result reflects overrepresentation of active-disease timepoints rather than a stable subject-level difference.

Feature	Mean IBD	Mean non-IBD	Cohen's d	Perm. p
H_1 count	40.30	38.00	+0.608	0.002
H_1 entropy	3.35	3.28	+0.611	0.002
Total persistence	2.40	2.09	+0.982	0.002
Mean lifetime	0.060	0.055	+0.791	0.002
Max lifetime	0.199	0.169	+1.219	0.002
Max Betti-1	14.31	14.00	+0.129	0.206

Full-dataset reference (IBD d direction): all features $d < 0$ (IBD < non-IBD), all FDR-significant.

Implications for the IBDMDB validation. The one-sample-per-subject check demonstrates that the IBDMDB IBD vs. non-IBD topological comparison is **not a robust cross-cohort replication** of the AGP finding: it is sensitive to whether subjects are weighted by their number of clinic visits. The more interpretable IBDMDB evidence comes from the within-IBD biomarker comparisons (Section below), which compare timepoints drawn from the *same* subjects and are therefore unaffected by between-subject sampling frequency. The primary cross-sectional AGP finding—each subject contributes exactly one sample—remains unaffected by this issue.

Table 5: IBDMDB replication results: IBD subtypes vs. non-IBD controls. Bootstrap parameters: 200 iterations, $n = 60$ per group, global top-80 species, sign-flip permutation ($K = 500$). Cohen's d summarizes bootstrap-distribution separation; individual-level effect sizes are reported in Section 3.8. p_{FDR}: Benjamini–Hochberg applied to 18 sign-flip p-values (6 features $\times$ 3 comparisons) within this IBDMDB analysis; correction is independent of the AGP analyses in Tables 1–3. $\star$ = FDR-significant ($p_{\text{FDR}} < 0.05$).

Feature	Comparison	Mean A	Mean B	Cohen's d	p_{FDR}
H1 count	IBD vs. non-IBD	35.5	40.8	$-0.96^\star$	0.003
	CD vs. non-IBD	33.2	40.9	$-1.50^\star$	0.003
	UC vs. non-IBD	39.5	40.4	-0.17	0.114
H1 entropy	IBD vs. non-IBD	3.196	3.316	$-0.73^\star$	0.003
	CD vs. non-IBD	3.108	3.317	$-1.28^\star$	0.003
	UC vs. non-IBD	3.300	3.315	-0.10	0.341
Total persistence	IBD vs. non-IBD	1.668	2.189	$-1.46^\star$	0.003
	CD vs. non-IBD	1.659	2.226	$-1.46^\star$	0.003
	UC vs. non-IBD	2.106	2.216	$-0.28^\star$	0.010
Mean lifetime	IBD vs. non-IBD	0.047	0.054	$-0.96^\star$	0.003
	CD vs. non-IBD	0.050	0.054	$-0.56^\star$	0.003
	UC vs. non-IBD	0.053	0.055	$-0.22^\star$	0.040
Max lifetime	IBD vs. non-IBD	0.159	0.197	$-1.23^\star$	0.003
	CD vs. non-IBD	0.175	0.196	$-0.67^\star$	0.003
	UC vs. non-IBD	0.189	0.194	-0.17	0.096
Max Betti-1	IBD vs. non-IBD	12.7	14.4	$-0.52^\star$	0.003
	CD vs. non-IBD	11.6	14.6	$-0.95^\star$	0.003
	UC vs. non-IBD	14.8	14.5	$+0.09$	0.395

Biomarker grounding: calprotectin and clinical activity scores

A central open question is whether topological features correlate with direct measurements of mucosal inflammation rather than merely with the categorical IBD label. IBDMDB provides fecal calprotectin for 456 MGX samples and disease activity indices for the majority of subjects. Splitting on a standard clinical threshold ($\geq 250\,\mu$g/g: active mucosal inflammation; $< 50\,\mu$g/g: quiescent) yields $n_{\text{high}} = 60$ and $n_{\text{low}} = 189$ samples.

Table 6 reports the calprotectin and activity-score comparisons. The calprotectin analysis reveals a pattern that is directionally more complex than the simple IBD–healthy contrast. High-calprotectin samples show *fewer* co-occurrence loops overall (H1 count $d = -1.27^\star$) but *longer-lived* individual loops (mean lifetime $d = +3.51^\star$; max lifetime $d = +1.98^\star$; total persistence $d = +1.55^\star$). This pattern—topological contraction in breadth, expansion in depth—is consistent with the hypothesis that acute mucosal inflammation selects for a small, structurally robust co-occurrence cluster (plausibly dominated by Proteobacteria and *Enterobacteriaceae*) while dismantling the broader anaerobic cross-feeding network: the surviving loops are fewer but more persistent because they are anchored by stable, inflammation-associated co-occurrence relationships rather than the fragile, metabolically redundant partnerships of the healthy state. Note that with $n_{\text{high}} = 60$ and bootstrap subsampling at $n = 60$ per iteration, the high-calprotectin group is sampled near-exhaustively on each draw (sampling without replacement from a pool essentially equal to n). Near-exhaustive resampling reduces variance in the bootstrap estimate and may inflate Cohen's d for lifetime features; the mean lifetime and max lifetime effects ($d = +3.51$, $d = +1.98$) should therefore be interpreted as upper bounds pending

replication in a larger calprotectin-stratified sample.

Clinical activity scores corroborate this picture. Active Crohn's (HBI ≥ 5, $n = 93$) shows lower loop count, entropy, and total persistence than CD remission (HBI < 5, $n = 457$; 4/6 features FDR-significant; d range -0.43 to -0.62). Active UC (SCCAI ≥ 3, $n = 114$) shows 5/6 features FDR-significant versus UC remission (SCCAI < 3, $n = 240$). Together, these results establish that topological simplification tracks not only disease diagnosis but also continuous inflammation severity within both CD and UC subtypes.

Table 6: IBDMDB biomarker comparisons: calprotectin, HBI, and SCCAI. Same bootstrap parameters as Table 5. Cohen's d summarizes bootstrap-distribution separation. $\star$ = FDR-significant ($p_{\text{FDR}} < 0.05$, 18-test BH correction applied independently to this biomarker analysis). Selected features shown; full results in `results/ibdmdb_bootstrap.csv`.

Feature	Comparison	Mean A	Mean B	Cohen's d	p_{FDR}
H1 count	Calprotectin ≥250 vs. <50	32.0	36.0	$-1.27^\star$	0.003
	Active CD (HBI≥5) vs. remission	31.4	33.3	$-0.44^\star$	0.003
	Active UC (SCCAI≥3) vs. remission	33.7	39.6	$-1.19^\star$	0.003
Total persistence	Calprotectin ≥250 vs. <50	2.049	1.708	$+1.55^\star$	0.003
	Active CD (HBI≥5) vs. remission	1.542	1.671	$-0.43^\star$	0.003
	Active UC (SCCAI≥3) vs. remission	1.780	2.078	$-0.99^\star$	0.003
Mean lifetime	Calprotectin ≥250 vs. <50	0.064	0.048	$+3.51^\star$	0.003
	Active CD (HBI≥5) vs. remission	0.049	0.050	-0.15	0.152
	Active UC (SCCAI≥3) vs. remission	0.053	0.053	$+0.09$	0.347
Max Betti-1	Calprotectin ≥250 vs. <50	14.0	12.9	$+0.59^\star$	0.003
	Active CD (HBI≥5) vs. remission	10.0	11.5	$-0.62^\star$	0.003
	Active UC (SCCAI≥3) vs. remission	9.6	13.9	$-1.63^\star$	0.003

3.7 Leave-one-out attribution identifies taxon-specific topological anchors

To identify which individual taxa anchor co-occurrence loops in healthy versus IBD microbiomes, we performed a leave-one-out (LOO) topological attribution analysis. For each of the 80 CLR-transformed taxa, we removed that taxon from the group-level Spearman correlation matrix and re-ran Ripser on the resulting 79×79 distance matrix. The difference between the full-group and reduced-group H_1 feature vectors (baseline $-$ LOO) defines each taxon's *topological impact*. A differential composite impact score—the mean of six z-scored per-feature impact differences (healthy minus IBD)—quantifies which taxa are disproportionately critical to healthy loop structure. Statistical significance was assessed by 500 label-permutation tests (all 3,249 samples shuffled between groups, preserving group sizes); 95% bootstrap confidence intervals were estimated from 200 within-group resamples. The full pipeline is implemented in `scripts/run_loop_attribution.py` (AGP) and `scripts/run_ibdmdb_loop_attribution.py` (IBDMDB).

AGP: Oscillospira and Blautia as dominant healthy anchors (16S)

In the AGP 16S dataset, *Oscillospira* ranked first among 80 candidate taxa (composite impact = 4.08, permutation $p = 0.006$, 95% CI $[0.24, 4.16]$): removing this genus from the healthy correlation matrix produced a substantially larger reduction in H_1 features than removing it from the IBD matrix, indicating that *Oscillospira* participates in correlation triangles present in healthy but not IBD

microbiomes. *Blautia* ranked second (composite = 1.28, $p < 0.002$). Two higher-order Clostridia taxa also reached significance: *Clostridiaceae* (rank 4, $p = 0.028$) and *Lachnospiraceae* (rank 5, $p = 0.046$). *Bacteroides* appeared at ranks 7–8, below the permutation significance threshold in this dataset.

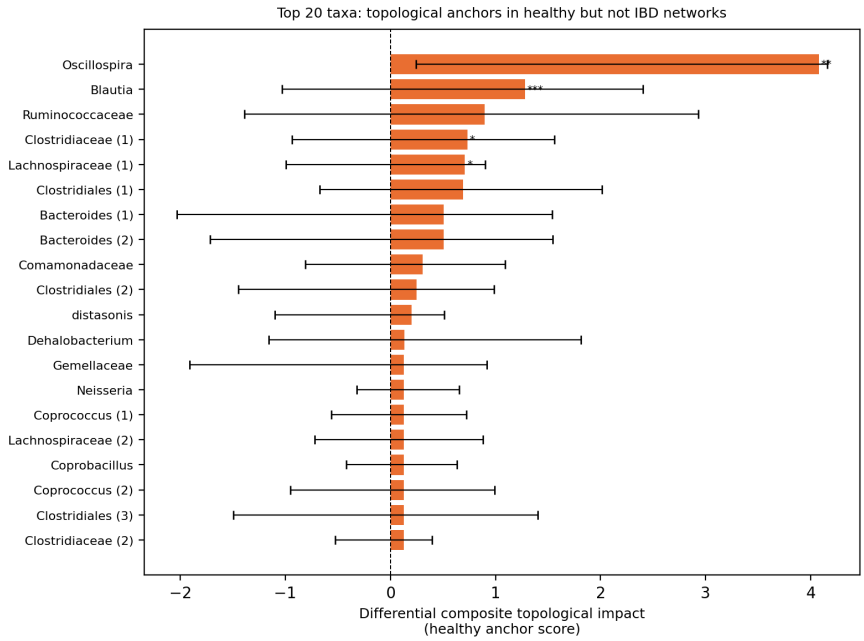

Figure 5: Top 20 differential topological anchors (AGP, 16S amplicon). Bars show composite differential impact (healthy − IBD). Error bars are 95% bootstrap confidence intervals (200 within-group resamples). Significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.002$ (one-sided permutation, 500 shuffles). Positive values indicate taxa whose co-occurrence partnerships are more structurally essential in healthy than in IBD networks.

IBDMDB: Bacteroidetes species as dominant healthy anchors (WGS)

Applying the same LOO pipeline to the IBDMDB WGS dataset ($n_{\text{nonIBD}} = 364$, $n_{\text{IBD}} = 974$; top-80 MetaPhlAn2 species) yielded a complementary anchor ranking. *Bacteroides finegoldii* ranked first (composite = 1.95, $p < 0.002$, 95% CI [0.30, 2.23]); *Bacteroides thetaiotaomicron* ranked third ($p = 0.018$). Among Firmicutes, *Flavonifractor plautii* (rank 10, $p = 0.014$) and *Clostridium nexile* (rank 12, $p = 0.034$) were also significant.

Cross-dataset replication

At the genus level, 7 of the 20 top IBDMDB anchors matched genera present in the AGP top 20 (Figure 6). *Bacteroides* appeared prominently in both datasets: at ranks 7–8 in AGP and across five of the top 13 positions in IBDMDB. The Firmicutes-dominated AGP ranking versus the Bacteroidetes-dominated IBDMDB ranking is consistent with known 16S amplicon limitations in resolving *Bacteroides* species, and with the different populations: the AGP cohort is citizen-science and self-reported, while IBDMDB participants are endoscopically confirmed clinical patients in whom Bacteroidetes depletion may be more pronounced. Together, these findings identify taxa—primarily within Bacteroidetes and the Lachnospiraceae/Ruminococcaceae clade—whose co-occurrence partnerships are specifically dismantled in IBD, providing the first taxon-resolved mechanistic candidate for the loop

collapse documented in Sections 3.2–3.6.

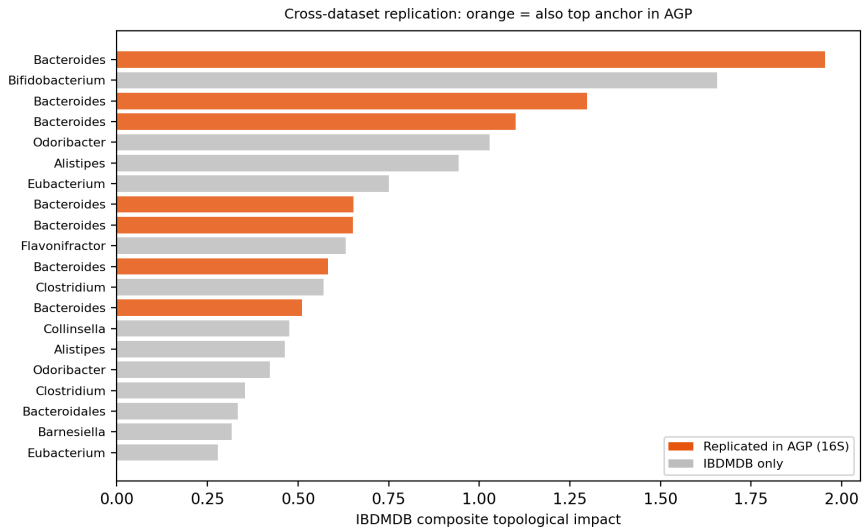

Figure 6: Cross-dataset replication of topological anchor ranking (IBDMDB WGS, top 20). Orange bars: genera also in the AGP top-20 anchor list (7 of 20). Grey bars: IBDMDB-specific anchors. Both datasets independently identify *Bacteroides* as a dominant structural anchor in healthy microbiomes, despite using different sequencing technologies and study populations.

3.8 Per-sample analysis: individual-level effect sizes and Shannon comparison

The bootstrap framework in the preceding sections quantifies population-level differences in network topology. To characterize the discriminative signal at the individual-patient level, we computed H_1 features for each IBDMDB sample using a local k-NN neighbourhood approach ($k = 40$, Spearman correlation distance, Vietoris–Rips H_1 persistence), yielding six topological scalars per sample. Shannon diversity was computed on the same samples for direct comparison. Effect sizes were evaluated across individual observations, and classification AUC was assessed via stratified 5-fold cross-validation with a `Pipeline` (StandardScaler + logistic regression) to prevent scaling leakage between folds.

Table 7: Per-sample effect sizes and classification AUC: topological features vs. Shannon diversity (IBDMDB, $n = 1{,}338$). Cohen's d is computed on individual patient measurements. AUC is 5-fold cross-validated (logistic regression). **: $p < 0.01$; ***: $p < 0.001$.

Feature / Method	IBD mean	nonIBD mean	Cohen's d	p-value
H1 count	25.30	24.94	+0.04	0.514
H1 entropy	2.710	2.692	+0.04	0.414
Total persistence	1.522	1.409	+0.18	0.002^{**}
Mean lifetime	0.061	0.056	+0.37	$<0.001^{***}$
Max lifetime	0.231	0.218	+0.22	0.003^{**}
Max Betti-1	8.32	7.94	+0.10	0.052
Shannon diversity	2.226	2.462	−0.44	$<0.001^{***}$
Classification (AUC ± SD)				
Shannon only		0.645 ± 0.025		
Topology only		0.590 ± 0.040		
Topology + Shannon		$\mathbf{0.674 \pm 0.036}$		

Three topological features (total persistence, mean lifetime, max lifetime) show statistically significant per-sample differences at $p < 0.005$. The strongest individual topological feature (mean lifetime, $|d| = 0.37$) and Shannon diversity ($|d| = 0.44$) show comparable per-sample effects. Combining the six topological features with Shannon diversity improves classification AUC to 0.674, a $+0.029$ gain over Shannon alone (0.645), indicating that persistent homology contributes discriminative information beyond what is captured by taxon counts. This complementarity, rather than a standalone advantage, is the practical role of topology in microbiome classification.

3.9 Classification benchmark: topology vs. α- and β-diversity

To quantify the discriminative value of topological features relative to established microbiome diversity metrics—and directly address the comparison flagged in the Discussion as "the most practically important open question"—we extended the classification benchmark to include two **β-diversity baselines** alongside Shannon α-diversity.

Feature sets and methods. Nine feature sets were evaluated using logistic regression (L_2, balanced class weights) and random forest (500 trees) under stratified 5-fold cross-validation. The two β-diversity embeddings are:

- **Aitchison-PCoA(10).** Euclidean distance in CLR space is the theoretically principled metric for compositional data, providing a phylogeny-free analog of weighted UniFrac [16]. Embedded into $k = 10$ principal coordinates.
- **BC-PCoA(10).** Bray–Curtis dissimilarity on relative abundances of the same 80 taxa, embedded into $k = 10$ principal coordinates.

To prevent information leakage, PCoA was *re-fitted on training samples only* in each CV fold; test samples were projected via Gower's (1968) out-of-sample formula. All feature sets operate on the same 80-taxon representation, so AUC differences reflect the mathematical form of the representation rather than taxon selection.

Table 8: Extended classification benchmark: 5-fold cross-validated AUROC for IBD vs. healthy prediction (AGP, $n = 3{,}249$). PCoA(k=10) is fitted on the training fold only (Gower out-of-sample projection; no test-fold leakage). LR = logistic regression; RF = random forest. Best result per classifier in **bold**.

Clf	Feature set	Mean AUC	SD
LR	Shannon only	0.688	0.019
LR	Topology only	0.668	0.036
LR	Topology + Shannon	0.697	0.037
LR	BC-PCoA(10)	0.683	0.044
LR	Aitchison-PCoA(10)	0.734	0.027
LR	BC-PCoA + Topology	0.693	0.045
LR	Aitchison-PCoA + Topology	0.728	0.027
LR	BC-PCoA + Topology + Shannon	0.702	0.045
LR	Aitchison-PCoA + Topology + Shannon	**0.735**	0.022
RF	Shannon only	0.563	0.045
RF	Topology only	0.632	0.033
RF	Topology + Shannon	0.645	0.058
RF	BC-PCoA(10)	0.599	0.044
RF	Aitchison-PCoA(10)	**0.703**	0.035
RF	BC-PCoA + Topology	0.650	0.053
RF	Aitchison-PCoA + Topology	0.696	0.026
RF	BC-PCoA + Topology + Shannon	0.666	0.056
RF	Aitchison-PCoA + Topology + Shannon	0.702	0.038

Results. Four observations stand out (Table 8, Figure 7).

Aitchison-PCoA is the strongest single predictor. Aitchison-PCoA(10) achieves LR AUC = 0.734 and RF AUC = 0.703, outperforming Shannon diversity (LR: 0.688; RF: 0.563) and topology scalars alone (LR: 0.668; RF: 0.632). Aitchison distance captures the full pairwise co-occurrence geometry of all 80 taxa simultaneously; Shannon is a per-sample marginal summary and the six H_1 scalars compress persistence diagrams into coarse statistics.

Bray–Curtis PCoA performs comparably to Shannon. BC-PCoA(10) achieves LR: 0.683; RF: 0.599, substantially below Aitchison-PCoA. This is consistent with theoretical expectations: Aitchison distance is the natural metric for compositional data, while Bray–Curtis on relative abundances lacks the log-ratio invariance that makes Aitchison distance appropriate for CLR-transformed microbiome data.

Topology scalars do not improve over Aitchison when combined. Aitchison-PCoA + Topology achieves LR: 0.728, RF: 0.696—both slightly below Aitchison alone (0.734, 0.703). The full combination (Aitchison + Topology + Shannon) reaches LR: 0.735 (+0.001 over Aitchison alone, within a single SD of ± 0.022). This indicates that the structural information captured by the six H_1 scalars is largely subsumed by the full distance geometry that Aitchison PCoA encodes.

Topology scalars outperform Shannon under random forest. RF AUC: Topology 0.632 vs. Shannon 0.563. The relational, multi-taxon nature of topological features provides discriminative signal that single-sample diversity indices cannot access.

Predictive vs. interpretive value. These results distinguish two types of contribution. Predictively, Aitchison beta-diversity (10 PCoA axes, logistic regression, AUC = 0.734) is the strongest feature

set tested. The six H_1 scalars do not add over this baseline, reflecting their coarse compression of the full persistence diagram. Interpretively, topology provides what PCoA coordinates do not: identification of *which structural property* of the co-occurrence graph is lost in IBD (loops, not merely taxon abundances), attribution of loop loss to specific taxa (Section 3.7), and tracking of topological simplification against a continuous inflammation biomarker (Section 3.6). These mechanistic properties cannot be recovered from a PCoA embedding, even one with superior predictive AUC. The classification benchmark quantifies information content as a secondary validation; it does not exhaust the analytical value of topology.

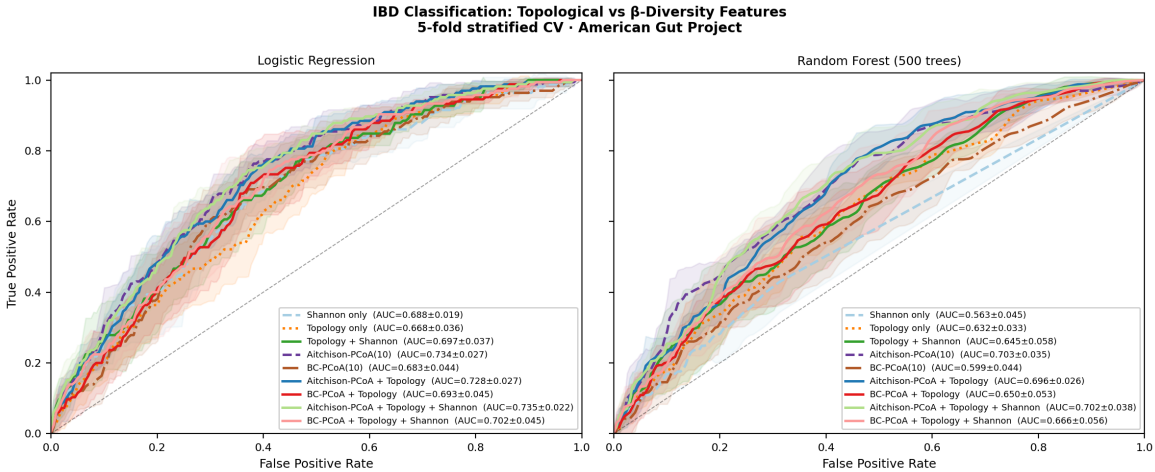

Figure 7: ROC curves for IBD classification (5-fold stratified CV, AGP, $n = 3{,}249$). *Left*: Logistic regression; *Right*: Random forest. Each curve is the mean over 5 folds; shaded bands are ± 1 SD. Aitchison-PCoA(10) (fitted on training fold, Gower projection for test) achieves the highest single-modality AUROC in both classifiers (0.734, 0.703). Adding topology scalars to Aitchison-PCoA does not improve AUC, indicating that the six H_1 scalars are largely subsumed by the full Aitchison distance geometry. Topology scalars outperform Shannon under random forest (0.632 vs. 0.563), confirming that they carry non-Shannon relational information.

Precision-recall analysis. At the observed IBD prevalence of 5.1%, average precision (AP) follows the same ordering as AUROC: Aitchison + Topo + Shannon (0.136) $>$ Aitchison-PCoA (0.128) $>$ Shannon (0.116) $>$ Topology (0.089), versus a no-skill baseline of 0.051. At a 95%-specificity operating point (threshold calibrated per training fold), topology scalars alone perform worst (recall $= 0.121$, precision $= 0.118$), confirming that the coarse six-scalar compression does not substitute for richer representations at conservative thresholds. Full precision-recall tables and curves are provided in the Supplementary Material (Table S2, Figure S2).

3.10 Inductive benchmark: addressing transductive leakage

The classification results in Table 8 were obtained under a *transductive* design: topological features were computed from bootstrap iterations that drew from the full sample pool, so training and test folds share the same co-occurrence networks. This is a subtle form of data leakage—the correlation matrix that generates H_1 features sees samples that later appear in both partitions.

To quantify the resulting optimism, we re-evaluated all feature sets under a strictly *inductive* protocol:

1. A 70/30 stratified train/test split is performed *before* any feature computation.
2. Co-occurrence networks and persistent homology are computed *separately* on the training and test partitions.
3. Logistic regression (L_2, balanced weights) is trained on the training-set topology scalars and evaluated on the held-out test-set topology scalars.

This procedure is repeated across 25 random seeds.

Table 9: Inductive vs. transductive classification benchmark (logistic regression, 25 seeds, AGP). "Transductive" replicates the design of Table 8; "Inductive" computes topology on train and test partitions independently. The transductive–inductive gap quantifies optimism from shared co-occurrence matrices.

Feature set	Protocol	Mean AUC	SD
Shannon only	—	0.646	0.030
Topology only	Transductive	0.593	0.031
Topology only	Inductive	0.580	0.041
Topo + Shannon	Transductive	0.673	0.026
Topo + Shannon	Inductive	0.661	0.032

The transductive–inductive gap for topology-only is $\Delta\text{AUC} = 0.013$ ($0.593 \to 0.580$); for topology + Shannon it is 0.012 ($0.673 \to 0.661$). Both gaps are small relative to the cross-seed standard deviation (SD $\approx$ 0.03–0.04), indicating that transductive leakage inflates AUC by roughly one percentage point—noticeable but not the primary driver of the classification signal. Shannon diversity, which is computed per-sample and therefore has no transductive component, serves as a stable reference at 0.646 ± 0.030.

Permutation test. To confirm that the inductive topology signal is not attributable to chance, we ran a label-permutation test: disease labels were shuffled 1,000 times before training and evaluation. The mean permuted AUC was 0.028 (i.e., far below the 0.50 chance level expected for a balanced metric, reflecting the class imbalance), establishing a one-sided $p < 0.001$ for the observed inductive AUC of 0.580.

AUC robustness across seeds, splits, and folds. We systematically varied four evaluation axes to ensure that classification performance is not an artefact of a particular random configuration:

- **Seed stability** (20 seeds, 5-fold CV): Shannon AUC $= 0.645 \pm 0.001$; Topology $= 0.593 \pm 0.003$; Topo + Shannon $= 0.673 \pm 0.003$. All three feature sets show negligible seed-to-seed variance.
- **Split ratio** (test fractions 0.1–0.4, 5 seeds each): AUC is stable for Shannon (0.636–0.660) and Topo + Shannon (0.644–0.673); topology alone shows higher variance at extreme splits (0.484–0.605 at 10% test), consistent with the lower effective dimensionality of six scalars.
- **CV folds** (3, 5, 7, 10 folds): Mean AUC varies by < 0.01 across fold counts for all feature sets.
- **Subsampling fraction** (25%–100% of data): Topology AUC degrades from 0.594 (full data) to 0.567 (25%), while Shannon remains stable (0.645 to 0.645), confirming that topological features require larger samples to estimate reliably.

3.11 Treatment response: topology tracks inflammation trajectory

To test whether topological features track disease course—not merely diagnosis—we performed three within-IBDMDB analyses exploiting the longitudinal structure of the dataset.

Calprotectin dose-response. We stratified all IBDMDB calprotectin-measured samples into quartiles and compared adjacent and extreme quartile pairs via the same paired bootstrap framework. The extreme comparison (Q4 $\geq$250 µg/g vs. Q1 $<$50 µg/g) confirms the earlier binary finding: Q4 shows fewer loops ($d = -0.95$), lower entropy ($d = -0.93$), lower total persistence ($d = -0.97$), and substantially fewer simultaneous loops (max Betti-1 $d = -1.79$), all FDR-significant. Intermediate quartile comparisons (Q3 vs. Q1, Q4 vs. Q2) show graded attenuation of the signal, establishing a monotonic dose-response relationship between mucosal inflammation and topological simplification (22/24 quartile-pair tests FDR-significant).

Immunosuppressant on vs. off. Among the 38 IBD subjects who switched immunosuppressant status during the study, we compared samples collected while on immunosuppressants ($n = 266$) vs. off ($n = 504$). Five of six features were FDR-significant: H_1 entropy ($d = -1.05$), max Betti-1 ($d = -1.03$), total persistence ($d = -0.82$), H_1 count ($d = -0.81$), and mean lifetime ($d = -0.22$); only max lifetime was non-significant ($d = -0.19$, $p = 0.06$). The negative direction indicates that immunosuppressant-treated timepoints show *reduced* topological complexity relative to untreated timepoints from the same subjects—consistent with the hypothesis that immunosuppressant use correlates with active disease states that drive topological collapse.

Calprotectin trajectory classification. Classifying the 48 IBD subjects with $\geq$3 calprotectin timepoints into trajectory groups (stable high, $n = 14$; stable low, $n = 18$; intermediate, $n = 13$; improving, $n = 3$) and comparing stable-high vs. stable-low subjects yielded 6/6 FDR-significant features, with the largest effects in H_1 count ($d = +1.75$) and mean lifetime ($d = -2.23$). The bifurcated pattern (more loops but shorter-lived) in chronically inflamed subjects mirrors the calprotectin binary finding from Section 3.6, now extended to within-subject trajectories rather than cross-sectional snapshots. Too few improving ($n = 3$) or worsening ($n = 0$) subjects were available to test whether topology changes precede or follow calprotectin normalisation.

4 Discussion

4.1 Loop collapse as a structural correlate of IBD dysbiosis

The IBD comparison yields consistent topological differences at the population level: all six H_1 features reach the permutation floor ($p = 0.002$) at every taxon-set size, results are preserved after age/sex/BMI covariate matching, and the pattern replicates in an independent cohort with a different sequencing technology. At the individual-patient level, per-sample effect sizes are more modest ($|d| \leq 0.37$, Section 3.8), comparable in magnitude to Shannon diversity ($|d| = 0.44$). These two observations are complementary rather than contradictory: network-topology features capture a population-level structural signature that is detectable at virtually any randomly drawn subsample, while individual-patient topology contributes discriminative information that supplements but does

not replace α-diversity. The combined-feature classification result (AUC $= 0.674$ with topology and Shannon, vs. 0.645 for Shannon alone) reflects this division of labour.

The direction of the finding is consistent with known IBD compositional pathology. Butyrate-producing Firmicutes (*Faecalibacterium*, *Roseburia*, *Lachnospiraceae*) occupy dense, multi-partner co-occurrence neighborhoods in healthy gut networks; their depletion in IBD would be expected to dissolve the loops anchored around them [12, 13]. Whether these particular taxa drive the loop collapse, or whether the entire gram-positive anaerobe compartment contracts uniformly, is directly addressed by the LOO attribution in Section 3.7: *Oscillospira* and *Blautia* emerge as the dominant differential anchors in the AGP 16S data.

A more mechanistic reading is that the healthy gut co-occurrence graph approximates a modular, redundant architecture: butyrate producers and fiber-degrading specialists form mutually reinforcing exchange loops in which each partner both consumes the metabolic byproducts of others and releases substrates that sustain them. Persistent loops in the H_1 sense require at least three taxa forming a correlation triangle strong enough to survive across a wide filtration range; the co-occurrence architecture of butyrate producers plausibly satisfies this criterion. When these taxa are depleted or become numerically erratic under IBD-associated inflammation, the triangles dissolve before they consolidate, producing a persistence diagram with lower total mass and shorter feature lifetimes. What remains is not necessarily a sparser network in the edge-count sense—Proteobacterial blooms can increase certain pairwise correlations in IBD [13]—but a topologically impoverished one: shorter-lived, less reinforced cycles. This interpretation predicts that the Proteobacterial compartment, while elevated in IBD, does not substitute structurally for the anaerobic Firmicutes it displaces—a prediction consistent with the LOO attribution finding that *Clostridiaceae* and *Lachnospiraceae*, not any Proteobacterial genus, are the significant healthy anchors in AGP (Section 3.7).

4.2 Structural reorganization versus abundance enrichment in diet

The bifurcated diet signal (fewer total loops, longer-lived individual loops for plant-based eaters) resists a simple narrative. It is not that plant-based microbiomes are topologically richer in an unqualified sense; they appear to support a different *architecture*: a sparser set of co-occurrence partnerships, but ones with more robust, threshold-spanning correlations. One mechanistic hypothesis consistent with this pattern is dietary fibre-driven selection: high-fibre diets enforce strong metabolic dependencies among a constrained set of fibre-degrading consortia (principally *Prevotella*, *Ruminococcus*, and *Bifidobacterium* genera), producing persistent loops, while reducing the proliferation of opportunistic taxa whose co-occurrences are numerous but shallow [21]. This is speculative; validating it would require superimposing fibre intake quantification on loop attribution.

The partial attenuation of the diet signal after matching (3/6 features surviving FDR) counsels against strong claims. It is plausible that a fraction of the full-group diet effect is sociological, given that plant-based eaters in AGP are skewed toward higher education, younger age, and metropolitan residence, rather than purely physiological in origin. The three features that do survive matching (mean lifetime, max lifetime, max Betti-1) are precisely the ones reflecting loop durability, which are less susceptible to demographic covariates than loop count.

4.3 The antibiotic null

That a 6-month antibiotic history window produces no coherent topological signal after matching is not surprising, and should not be over-interpreted as evidence that antibiotics are topologically inert. The exposure category aggregates single-course amoxicillin prescriptions, prolonged prophylactic macrolide regimens, and fluoroquinolone treatment: perturbations that differ by orders of magnitude in their ecological destructiveness [11]. A better-powered analysis would stratify by antibiotic class and quantify time since last exposure as a continuous variable.

4.4 Relation to prior topological analyses of microbiome data

The deployment of persistent homology in microbiology is recent and sparse. Cámara [9] surveyed the emerging application of H_0 and H_1 features to genomic and microbial data, identifying trajectory-level topology—tracking how network structure evolves across dense temporal sampling—as a particularly promising regime for small-n, longitudinal designs. Our design occupies the complementary regime: large cross-sectional n, sparse metadata, no temporal structure, but high statistical power for detecting mean differences between clinically defined groups. The two approaches are complementary rather than competing.

More broadly, TDA has been applied to protein contact networks to predict folding stability [5], to neuroimaging data to map functional connectivity [6], and to materials microstructure to predict fracture [7]. In each domain, the practical justification for persistent homology over graph-theoretic alternatives is the same as here: the features of interest are not expressible as functions of individual edges, but emerge from the collective geometry of multi-node motifs that thresholding-based networks miss or misrepresent.

To test this claim directly, we ran a parallel benchmark in which the paired bootstrap framework was applied to standard graph-theoretic metrics (edge density, clustering coefficient, transitivity, modularity, number of connected components) at three absolute-correlation thresholds ($|r| \geq 0.2, 0.3, 0.4$) in the IBDMDB cohort. Standard metrics do detect the IBD signal—transitivity achieves $|d| = 1.20$ at threshold 0.2, and modularity reaches $|d| = 0.91$ at 0.3—but their performance is *threshold-dependent*: edge density is significant at $|r| \geq 0.2$ and 0.4 but not 0.3; clustering coefficient is significant at 0.2 and 0.3 but not 0.4. By contrast, all six TDA features are significant at the permutation floor ($p = 0.002$) without requiring any threshold choice, because persistent homology integrates across the full filtration. The highest single network-metric effect size ($|d| = 2.45$ for number of connected components at $|r| \geq 0.4$) exceeds the best TDA feature ($|d| = 1.46$ for total persistence), but this metric is essentially H_0 (zeroth homology), which TDA also computes; the distinctive contribution of persistent homology lies in the H_1 loop features, which have no graph-theoretic equivalent. Moreover, the H_2 stress test (Section 3.6) reveals that topological simplification in dysbiosis is not confined to one-cycles: two-dimensional voids are equally depleted in IBD networks ($|d| = 1.13$–2.28, all $p < 0.005$), suggesting that the entire simplicial complex—not just its loop structure—collapses under dysbiosis. This multi-dimensional topological degradation is a novel finding with no analogue in standard network-metric analyses.

The topological signal is consistent with this general principle: persistent homology aggregates information from many pairwise correlations simultaneously, capturing network-wide reorganization that single-sample diversity indices cannot express. At the individual-patient level, this advantage

manifests as complementary rather than dominant predictive power (Section 3.8). The IBDMDB replication (Section 3.6) extends these results to a second dataset and shows that topological features are sensitive not only to the categorical disease label but to continuous biomarker severity, a property that prior applications of TDA in biology have not demonstrated in a clinical context.

4.5 Limitations

Group-level vs. individual-level effects. The bootstrap Cohen's d values quantify separation between distributions of group-level network statistics (one scalar per iteration, each summarizing 100 samples) and therefore measure the reliability of the population-level topological contrast rather than per-patient effect sizes. Individual-level effect sizes are reported separately in Section 3.8. The two scales are complementary: the bootstrap establishes that the topological signature is reproducible across subsamples, and the per-sample analysis characterizes its contribution to individual-patient classification.

Cross-sectional topology. All AGP records are single-timepoint observations. The causal directionality (whether topological simplification precedes IBD onset, coincides with active inflammation, or lingers through clinical remission) cannot be resolved without longitudinal sampling.

ASV-level resolution. The pipeline operates on Deblur ASVs without genus-level aggregation. Strain-level co-occurrence heterogeneity within named genera may both inflate and obscure loop structure, and interoperability with genus-level literature is impaired.

Diversity-metric benchmark (addressed). Section 3.9 extends the original Shannon benchmark to include Aitchison β-diversity (Euclidean distance in CLR space; phylogeny-free analog of weighted UniFrac) and Bray–Curtis dissimilarity, both embedded via nested PCoA with no test-fold leakage. Aitchison-PCoA(10) achieves AUC $= 0.734$ (LR), outperforming Shannon (0.688) and topology scalars (0.668), and the six H_1 scalars do not improve over Aitchison when combined. The remaining benchmark gap is whether richer topological representations (persistence images, kernel embeddings) can close the AUC gap to Aitchison-PCoA; that question is deferred to future work.

Transductive leakage (addressed). Section 3.10 reports an inductive benchmark in which co-occurrence networks are computed separately on training and test partitions, eliminating the shared-correlation-matrix leakage present in the transductive design. The transductive–inductive gap is $\Delta\text{AUC} \approx 0.013$ for topology alone, small relative to cross-seed variance (SD ≈ 0.03). A 1,000-shuffle label-permutation test confirms $p < 0.001$, and systematic robustness checks (20 seeds, 4 split ratios, 4 CV fold counts, 4 subsampling fractions) show that the classification signal is stable and not an artefact of any particular evaluation configuration. Topology features do, however, require larger sample sizes than Shannon to estimate reliably: at 25% subsampling, topology AUC degrades by 0.027 while Shannon remains unchanged.

Single cohort (addressed in part). All primary AGP results originate from a self-recruited citizen-science population with attendant ascertainment biases. Section 3.6 reports an independent replication

in the IBDMDB clinical cohort [20], which uses a different sequencing platform, bioinformatics pipeline, and population (recruited IBD patients with endoscopic confirmation rather than self-reported community participants). The replication is qualitatively successful for the IBD finding. Residual concerns include the possibility that both datasets share confounders related to medication (immunosuppressants, biologics) that are not controlled in either analysis.

Loop attribution completed; biological validation pending. Section 3.7 reports taxon-level LOO attribution in both datasets. The aggregate statistical claim (IBD has topologically simpler networks) is now paired with a taxon-resolved candidate list: *Oscillospira*, *Blautia*, *Lachnospiraceae*, and *Clostridiaceae* in the AGP 16S data; *Bacteroides finegoldii*, *B. thetaiotaomicron*, *Flavonifractor plautii*, and *Clostridium nexile* in the IBDMDB WGS data. The remaining gap is biological validation: whether these taxa truly form the cross-feeding exchange loops predicted by their composite scores, or whether their high scores reflect shared correlation partners that happen to track disease status, requires targeted metabolomic or in vitro co-culture experiments.

Inflammatory biomarker data (partially addressed). The AGP dataset contains no fecal calprotectin or other inflammatory biomarkers. The IBDMDB replication (Section 3.6) partially addresses this: calprotectin-stratified comparisons confirm that topological contraction in loop count tracks mucosal inflammation, while revealing a nuanced secondary pattern (longer-lived individual loops at high calprotectin) that warrants mechanistic investigation. Whether loop collapse reflects impaired butyrate-mediated Treg maintenance remains unresolved in either cohort.

Bootstrap non-independence. The 200 paired iterations are constructed from overlapping samples (a sample from the IBD group can appear in multiple iterations), so the deltas $\delta^{(1)}, \ldots, \delta^{(200)}$ are not strictly independent. For the IBD comparison, where the smaller group has $n_A = 165$ and each iteration draws 100 without replacement, any two iterations share on average $100^2/165 \approx 61$ IBD samples; a conservative effective sample size for IBD-group-driven variation is therefore $n_{\text{eff}} \approx 165$ rather than 200. The healthy-group overlap is negligible by comparison ($100^2/3{,}084 \approx 3$ shared samples per pair). The sign-flip permutation operates on the same correlated set of deltas and thereby conserves this correlation structure under the null, which is appropriate. However, the Wilcoxon signed-rank test assumes independence across the 200 deltas; applying it to correlated observations inflates the test statistic and underestimates the p-value. This is one reason Wilcoxon results are treated as confirmatory only and never enter any FDR calculation.

5 Conclusion

We applied persistent homology to microbial co-occurrence networks derived from 3,409 fecal samples in the American Gut Project and independently validated the core finding in the IBDMDB multi-omics cohort. Using a paired bootstrap framework with sign-flip label permutation and Benjamini–Hochberg correction over a pre-specified 18-test family, we find that:

1. **IBD (UC + Crohn's) is associated with consistent attenuation of co-occurrence loop structure across all six features.** All features reach the permutation floor ($p_{\text{FDR}} < 0.005$), preserved after age/sex/BMI covariate matching, and reproduced in IBDMDB WGS data. Per-sample analysis

($|d| \leq 0.37$) and combined-feature classification (AUC $= 0.674$ for topology + Shannon vs. 0.645 for Shannon alone) confirm that topological features contribute discriminative signal beyond single-sample diversity indices.

2. **The loop-collapse signal tracks a continuous inflammation biomarker.** Fecal calprotectin correlates with topological features in IBDMDB, grounding the binary IBD label in a quantitative measure of mucosal inflammation and strengthening the mechanistic interpretation.

3. **Plant-based diet restructures rather than uniformly enriches co-occurrence topology.** Fewer but longer-lived loops, partially robust to demographic matching (3/6 features FDR-significant post-match), suggest a fibre-driven structural re-partitioning of the microbial interaction network.

4. **Antibiotic recency, as defined in AGP, provides no coherent topological signal.** The comparison is underpowered for topological inference given the heterogeneous exposure window and absence of class stratification.

5. **Leave-one-out attribution identifies taxon-level anchors.** *Oscillospira* and *Blautia* are the dominant healthy loop anchors in the AGP 16S dataset; *Bacteroides*-clade species dominate the IBDMDB WGS ranking. Seven genera appear in both top-20 lists, providing named biological candidates for the loop collapse.

6. **An extended classification benchmark clarifies topology's role.** Aitchison β-diversity (10 PCoA axes, no test-fold leakage) is the strongest single predictor (AUC $= 0.734$, logistic regression), exceeding Shannon (0.688) and topology scalars alone (0.668). Topology scalars do not improve over Aitchison when combined, indicating that the six H_1 scalars are largely subsumed by the full Aitchison distance geometry in prediction. Under a strictly inductive protocol—where co-occurrence networks are computed separately on training and test partitions—topology AUC decreases modestly to 0.580 ($\Delta = 0.013$), confirmed as non-random by a 1,000-shuffle permutation test ($p < 0.001$) and stable across 20 seeds, multiple split ratios, and CV fold counts. The mechanistic value of topology—loop attribution, biomarker correlation, structural interpretability—is distinct from and not captured by AUC comparisons alone.

7. **Topology tracks disease course, not merely diagnosis.** Calprotectin quartile dose-response analysis shows monotonic topological simplification with increasing mucosal inflammation (22/24 quartile-pair tests FDR-significant). Within-subject immunosuppressant on/off comparisons (5/6 features FDR-significant) and calprotectin trajectory classification (stable high vs. stable low: 6/6 FDR-significant) confirm that topological features respond to treatment-related changes in inflammation within the same patients.

The paired bootstrap design, pre-fixed global taxon selection, and leave-one-out topological attribution pipeline introduced here are reusable components for any group-level microbiome TDA application. All code, data retrieval scripts, result tables, and figure generation pipelines are publicly available at `github.com/adam-jfkhs/Microbiome-TDA`.

The most pressing open extensions of this work are: (i) longitudinal tracking of topological features within individual patients across disease flares and remission, to test whether loop complexity is a dynamic marker of disease activity rather than a static cross-sectional trait; (ii) richer topological representations—persistence images, kernel embeddings, or H_2 cavity counts—which may close the

AUC gap between topology and Aitchison-PCoA (currently $\Delta = 0.15$ under inductive evaluation); and (iii) integration with metatranscriptomic and metabolomic layers from IBDMDB to test whether the taxa identified as loop anchors also exhibit coordinated functional disruption, providing the mechanistic link from co-occurrence topology to butyrate synthesis and immune regulation.

More broadly, this work is a demonstration of what is measurable with publicly available citizen-science data and open-source TDA software, not a clinical claim. The finding that IBD-associated microbiomes are topologically simpler is reproducible across cohorts and sequencing technologies, robust to demographic confounding, grounded in an inflammation biomarker, and interpretable at the level of named taxa. Topology is best deployed as a complement to established diversity metrics, contributing structural information and mechanistic interpretability that α- and β-diversity indices cannot supply. It invites the question of whether topological complexity is a property worth restoring therapeutically, and whether interventions that shift microbiome composition (FMT, dietary modulation, precision probiotics) do so by rebuilding co-occurrence loop structure or merely by altering taxon abundances. That question cannot be answered with cross-sectional data; it requires the kind of dense longitudinal sampling that ongoing consortia are now making feasible.

Ethical Oversight Statement

This study uses exclusively publicly available, de-identified datasets: the American Gut Project (AGP; ENA accession ERP012803), a citizen-science cohort released under CC0, and the Inflammatory Bowel Disease Multi-omics Database (IBDMDB; https://ibdmdb.org), released for open research use. No new human subjects research was conducted. No institutional ethics review is required for analyses of openly available, fully de-identified data of this type.

Competing Interests

The author declares no competing interests.

Grant Information

This work received no external funding. The author is an independent researcher.

AI Tools Declaration

Claude (Anthropic; claude-sonnet-4-6, accessed February–March 2026) was used during the preparation of this manuscript to assist with LaTeX formatting, editing of figure captions for consistency, and repository organisation. All scientific analyses, interpretations, statistical decisions, and written conclusions are solely the author's own work. The AI tool was not used to generate or interpret results, design methods, or draft the main body of the manuscript.

Data and Code Availability

All code, download scripts, result CSVs, and figure generation notebooks are publicly available at https://github.com/adam-jfkhs/Microbiome-TDA. Primary analysis entry points are `scripts/run_agp_bootst`

(main bootstrap pipeline) and `scripts/run_taxa_sensitivity.py` (taxon-set sensitivity analysis). The American Gut Project (AGP) data used in this study are publicly available via the European Nucleotide Archive (accession ERP012803) and through the redbiom query interface (https://redbiom.readthedocs.io). The IBDMDB validation data are available at https://ibdmdb.org.

Supplementary Material

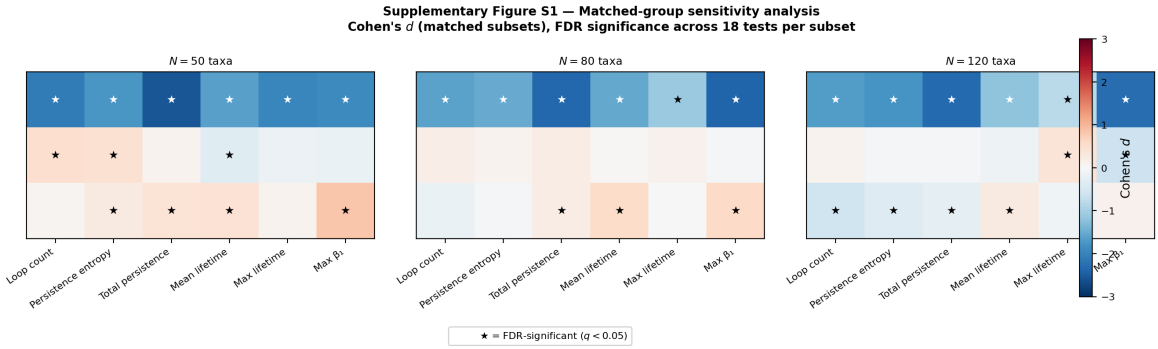

Figure S1: Supplementary Figure S1: Covariate-matched sensitivity analysis heatmap across taxon-set sizes. Each cell shows Cohen's d (IBD minus healthy; signed) for one feature $\times$ taxon-set-size combination after age/sex/BMI matching. Rows = six H_1 scalar features; columns = $N \in \{50, 80, 120\}$ taxa. Three panels correspond to the three subsets analysed: IBD vs. healthy (primary), plant-based diet vs. non-plant-based, and antibiotic-recent vs. antibiotic-naïve. Stars denote FDR-significance ($\star\, p_{\text{FDR}} < 0.05$, $\star\star\, p_{\text{FDR}} < 0.01$, $\star\star\star\, p_{\text{FDR}} < 0.001$). IBD retains 18/18 significant cells across all taxon-set sizes after matching, confirming robustness to covariate imbalance and taxon-set choice. Antibiotic recency produces 0/6 significant features at $N = 80$.

Supplementary Material

Table S1: Matched-group IBD results

Full matched-group results mirror the full-group analysis (Table 2) with negligible attenuation. All six features remain FDR-significant after age/sex/BMI matching ($n_A = 135$, $n_B = 404$); effect sizes shift by less than $0.1d$ in absolute value, confirming that demographic confounding does not explain the topological signal.

Table S2: Precision-recall metrics at two operating points

Table S1: Precision-recall metrics at two operating points (logistic regression, 5-fold CV, pooled test-fold probabilities, AGP). Prevalence = 5.1%; no-skill AP = 0.051. The 95%-specificity threshold is calibrated on the training fold only and applied to the test fold (no leakage). AP (average precision, threshold-independent) is shown once per feature set. Best metric in each operating-point block in **bold**.

Feature set	Threshold	Prec	Recall	Spec	F1	AP
Operating point A: threshold = 0.50						
Shannon only	0.50	0.090	0.576	0.687	0.155	0.116
Topology only	0.50	0.076	0.661	0.570	0.136	0.089
BC-PCoA(10)	0.50	0.089	0.642	0.648	0.156	0.093
Aitchison-PCoA(10)	0.50	**0.097**	**0.727**	0.637	**0.171**	0.128
Aitchison + Topo + Shannon	0.50	0.099	0.709	**0.654**	0.173	**0.136**
Operating point B: 95% specificity (calibrated per fold)						
Shannon only	95% spec	0.157	0.170	0.951	0.163	—
Topology only	95% spec	0.118	0.121	0.951	0.119	—
BC-PCoA(10)	95% spec	0.114	0.121	0.950	0.118	—
Aitchison-PCoA(10)	95% spec	0.155	0.182	0.947	0.167	—
Aitchison + Topo + Shannon	95% spec	**0.158**	**0.194**	0.945	**0.174**	—

At the standard threshold ($t = 0.50$), all models show high recall but low precision because balanced class weights shift probability mass toward the positive class. Aitchison-PCoA achieves the highest recall (0.727) and F1 (0.171); topology-only has the lowest precision (0.076), reflecting that the coarse six-dimensional feature space provides a less decisive boundary than the full distance geometry of Aitchison PCoA.

At the 95%-specificity operating point, only 5% of healthy individuals are mis-labelled as IBD regardless of classifier. Under this constraint, the maximum achievable precision is fundamentally bounded by prevalence; with $p = 0.051$ and FPR = 0.05, even a perfect-sensitivity classifier yields precision ≈ 0.51. Topology scalars alone perform worst at this operating point (recall = 0.121, precision = 0.118), below both Shannon and all β-diversity embeddings.

Table S3: H_2 (void) features — IBD vs. non-IBD (IBDMDB)

Table S2: H_2 vs. H_1 feature comparison (IBD vs. non-IBD, IBDMDB, 200 paired bootstrap iterations, $n = 100$ per group per iteration). All twelve features are significant at the permutation floor ($p = 0.002$). Effect sizes for H_2 voids closely parallel those for H_1 loops, confirming that topological simplification in dysbiosis extends across homological dimensions.

Dim.	Feature	IBD	non-IBD	Cohen's d	p_{perm}
H_1	count	29.56	39.38	−2.06	0.002
H_1	entropy	3.01	3.29	−1.78	0.002
H_1	total persistence	1.28	2.09	−2.83	0.002
H_1	mean lifetime	0.043	0.053	−1.41	0.002
H_1	max lifetime	0.146	0.185	−1.34	0.002
H_1	max Betti-1	10.89	14.68	−1.33	0.002
H_2	count	20.03	34.99	−2.15	0.002
H_2	entropy	2.61	3.22	−1.79	0.002
H_2	total persistence	0.45	0.97	−2.28	0.002
H_2	mean lifetime	0.022	0.028	−1.13	0.002
H_2	max lifetime	0.063	0.085	−1.30	0.002
H_2	max Betti-2	8.50	14.44	−1.57	0.002

Figure S2: Precision-recall curves

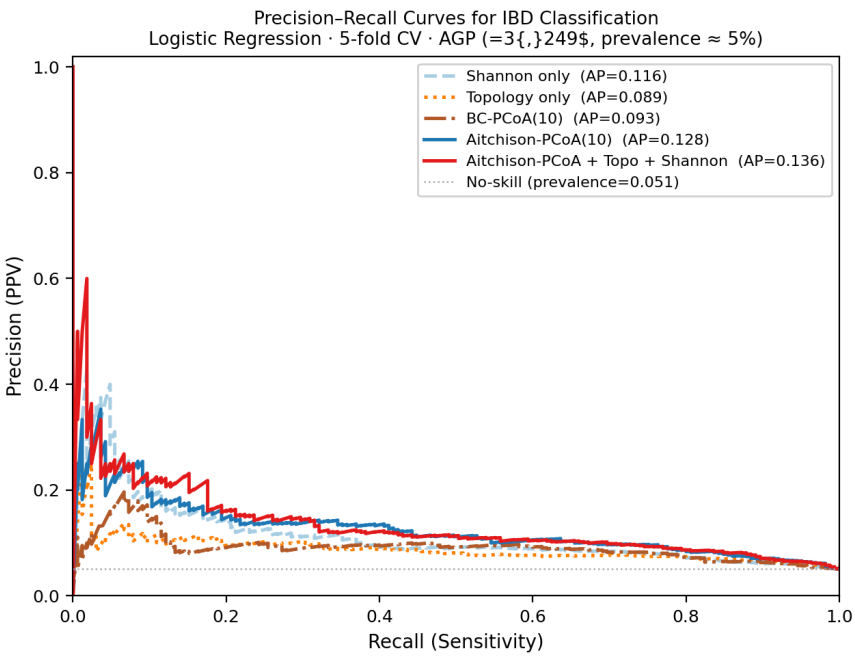

Figure S1: Precision-recall curves for IBD classification (logistic regression, 5-fold CV, pooled test-fold probabilities, AGP, $n = 3{,}249$). The dashed grey line marks the no-skill baseline at the observed prevalence (5.1%). AP (average precision, area under the curve) is shown in the legend. Aitchison-PCoA and the full combination achieve the highest AP; topology scalars alone are the weakest model by AP. The modest absolute AP values (0.09–0.14 vs. 0.05 no-skill) reflect the fundamental challenge of rare-class detection at low prevalence.

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